PSEUDO-HYPERJUMP INVERSION FAILS FOR TURING DEGREES

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Abstract. Jananthan and Simpson asked in Conjecture 4.5 whether every pseudo-hyperjump $\mathrm{HJ}_e(X) = X \oplus V_e^X$, with V_e^X uniformly $\Pi_1^{1,X}$, admits Turing-degree inversion above Kleene's $\mathcal{O}$. We settle their conjecture in the negative. We construct one index e_* such that $X <_{\mathrm{T}} \mathrm{HJ}_{e_*}(X)$ for every real X, while the range of HJ_{e_*} omits every Turing degree in the interval

$$\{\mathbf{d} : \deg_{\mathrm{T}}(\mathcal{O}) \leq \mathbf{d} < \deg_{\mathrm{T}}(\mathcal{O}^{\mathcal{O}})\}.$$

Thus Conjecture 4.5 fails even for an operator which strictly raises the Turing degree of every input. Each of Properties 4.6–4.8 implies Conjecture 4.5; consequently, the accompanying class-characterisation problems collapse: no class of reals, even without a definability assumption, satisfies any one of these properties.

1. Introduction

We identify reals with subsets of $\mathbb{N}$ (equivalently, with elements of $2^{\mathbb{N}}$), and use $\leq_{\mathrm{T}}$, $\leq_{\mathrm{HYP}}$, and $\leq_m$ for Turing, hyperarithmetical, and many-one reducibility, respectively. The corresponding strict and equivalence relations are denoted in the usual way. The Turing jump of X is denoted by X'. The ordinal ω_1^X is the least ordinal which is not the order type of an X-recursive well-ordering, and $\omega_1^{\mathrm{CK}} = \omega_1^{\emptyset}$ is the Church–Kleene ordinal, the least noncomputable ordinal. All analytical pointclasses are lightface, with any oracle parameter indicated explicitly. We write $\deg_{\mathrm{T}}(X)$ for the Turing degree of X, and $X \oplus Y$ for the usual effective join of two reals.

For a real X, let $\mathcal{O}^X$ denote the relativized Kleene $\mathcal{O}$, the hyperjump of X. It is $\Pi_1^{1,X}$-complete, uniformly in X. Equivalently, its Turing degree is that of the set of indices of X-recursive well-orderings. We put $\mathcal{O} = \mathcal{O}^{\emptyset}$.

Let $(V_e^X)_{e\in\mathbb{N}}$ be a standard effective enumeration, uniform in X, of the $\Pi_1^{1,X}$ subsets of $\mathbb{N}$. Jananthan and Simpson [1] define

$$\mathrm{HJ}_e(X) = X \oplus V_e^X$$

and ask whether the following analogue of the pseudojump inversion theorem holds [1, Conjecture 4.5]:

$$\forall e\, \forall A \geq_{\mathrm{T}} \mathcal{O}\, \exists B \quad A \equiv_{\mathrm{T}} \mathrm{HJ}_e(B) \equiv_{\mathrm{T}} B \oplus \mathcal{O}.$$

Here and below $A \geq_{\mathrm{T}} \mathcal{O}$ abbreviates $\mathcal{O} \leq_{\mathrm{T}} A$. We settle Conjecture 4.5 in the negative. More precisely, we construct one index e_* whose pseudo-hyperjump strictly raises every input degree but omits every degree in

$$\{\mathbf{d} : \deg_{\mathrm{T}}(\mathcal{O}) \leq \mathbf{d} < \deg_{\mathrm{T}}(\mathcal{O}^{\mathcal{O}})\}.$$

Immediately after stating Conjecture 4.5, Jananthan and Simpson observe that even an affirmative answer would leave open the characterisation of the Σ_1^1 classes satisfying their Properties 4.6–4.8. Our negative answer has the opposite consequence: as we observe in Section 3, each of Properties 4.6–4.8 implies Conjecture 4.5, and hence none of these properties is satisfied by any class.

The construction uses the change in behaviour at the predicate $\omega_1^X = \omega_1^{\mathrm{CK}}$. The class of reals satisfying this predicate is Σ_1^1; therefore its complement can be used as a Π_1^1 switch inside a single pseudo-hyperjump. References to the conjecture and numbered properties of [1] use

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arXiv:2101.08818v2 (19 July 2024). The inversion question already appears in Simpson's 2019 NUS lecture, reporting joint work with Jananthan [3, slide 9].

We use the following standard facts about the hyperjump:

(i) $X <_{\mathrm{HYP}} \mathcal{O}^X$; this follows from the completeness of $\mathcal{O}^X$ and [2, Theorem II.7.6(i), p. 51].
(ii) If $Y \leq_{\mathrm{HYP}} X$, then $\mathcal{O}^Y \leq_{\mathrm{T}} \mathcal{O}^X$; in fact, $\mathcal{O}^Y \leq_m \mathcal{O}^X$ [2, Proposition II.7.1, p. 49].
(iii) $\omega_1^X > \omega_1^{\mathrm{CK}}$ if and only if $\mathcal{O} \leq_{\mathrm{HYP}} X$ [2, Corollary II.7.7, p. 51].

The uniform completeness of the usual hyperjump also gives $X \oplus X' \leq_{\mathrm{T}} \mathcal{O}^X$ uniformly in X [2, Subsection II.5.4, p. 43].

2. A pseudo-hyperjump with a gap

The construction uses two columns with different roles. The even column uniformly codes X', ensuring strict Turing-degree increase, while the odd column codes $\mathcal{O}^X$ precisely when $\omega_1^X > \omega_1^{\mathrm{CK}}$. We first verify that the switch between these two cases has the required effective descriptive complexity.

Lemma 2.1. *The class*

$$\mathcal{L} = \{X : \omega_1^X = \omega_1^{\mathrm{CK}}\}$$

is Σ^1_1 [2, Corollary III.1.5, p. 54]. *Consequently, the predicate*

$$\omega_1^X > \omega_1^{\mathrm{CK}}$$

is a Π^1_1 *predicate of* X.

Proof. Let $(\varphi_e^X)_{e\in\mathbb{N}}$ be a uniform effective enumeration of the partial X-recursive functions from $\mathbb{N}^2$ to $\{0,1\}$, viewed as putative characteristic functions of strict linear orderings on $\mathbb{N}$. Write R_e^X for the corresponding candidate presentation, and let $R_i = R_i^{\emptyset}$.

Since $\omega_1^{\mathrm{CK}} \leq \omega_1^X$ for every X, we have $\omega_1^X = \omega_1^{\mathrm{CK}}$ if and only if, for every e, either R_e^X does not present a well-ordering or there are i and a real f such that R_i is a total linear ordering and f is an isomorphism from R_i onto R_e^X. Indeed, in the equality case every X-recursive well-ordering has recursive order type. If $\omega_1^X > \omega_1^{\mathrm{CK}}$, an X-recursive presentation of ω_1^{CK} has no recursive copy.

Failure to present a well-ordering is a $\Sigma^1_1(X)$ condition: failure to be a total linear ordering is arithmetical in X, while ill-foundedness is witnessed by an infinite descending sequence. The existence of i and of the isomorphism f is also $\Sigma^1_1(X)$. Finally, the universal number quantifier over e can be absorbed into a single existentially quantified real coding a sequence of tagged witnesses. The resulting definition is uniform and effective, so the class is Σ^1_1. □

Definition 2.2. Uniformly in X, define $V_*^X \subseteq \mathbb{N}$ by

$$2n \in V_*^X \iff n \in X',$$
$$2n+1 \in V_*^X \iff \omega_1^X > \omega_1^{\mathrm{CK}} \ \&\ n \in \mathcal{O}^X.$$

The first clause is arithmetical in X, and hence can be written as a $\Pi^{1,X}_1$ predicate. By Lemma 2.1, the second clause is the conjunction of two $\Pi^{1,X}_1$ predicates. Hence, using the arithmetical parity distinction, V_*^X is uniformly $\Pi^{1,X}_1$. Fix an index e_* such that $V_{e_*}^X = V_*^X$ for every real X.

Theorem 2.3 (Gap theorem)**.** *For every real* X,

$$X <_{\mathrm{T}} \mathrm{HJ}_{e_*}(X).$$

Moreover, the following dichotomy holds:

$$\omega_1^X = \omega_1^{\mathrm{CK}} \implies \mathrm{HJ}_{e_*}(X) \equiv_{\mathrm{T}} X' \text{ and } \mathcal{O} \not\leq_{\mathrm{T}} \mathrm{HJ}_{e_*}(X),$$
$$\omega_1^X > \omega_1^{\mathrm{CK}} \implies \mathrm{HJ}_{e_*}(X) \equiv_{\mathrm{T}} \mathcal{O}^X \text{ and } \mathcal{O}^{\mathcal{O}} \leq_{\mathrm{T}} \mathrm{HJ}_{e_*}(X).$$

Consequently,

$$\{\deg_{\mathrm{T}}(\mathrm{HJ}_{e_*}(X)) : X \in 2^{\mathbb{N}}\} \cap \{\mathbf{d} : \deg_{\mathrm{T}}(\mathcal{O}) \leq \mathbf{d} < \deg_{\mathrm{T}}(\mathcal{O}^{\mathcal{O}})\} = \varnothing.$$

Proof. The even column gives $X' \leq_{\mathrm{T}} \mathrm{HJ}_{e_*}(X)$ for every X. Thus $X <_{\mathrm{T}} X' \leq_{\mathrm{T}} \mathrm{HJ}_{e_*}(X)$, proving strictness.

Suppose first that $\omega_1^X = \omega_1^{\mathrm{CK}}$. The odd column in Definition 2.2 is empty, so

$$\mathrm{HJ}_{e_*}(X) = X \oplus V_*^X \equiv_{\mathrm{T}} X'.$$

Spector's criterion gives $\mathcal{O} \not\leq_{\mathrm{HYP}} X$. Since $X' \leq_{\mathrm{HYP}} X$, we have $\mathcal{O} \not\leq_{\mathrm{T}} X'$; otherwise $\mathcal{O} \leq_{\mathrm{T}} X' \leq_{\mathrm{HYP}} X$, a contradiction. This proves the first alternative.

Now suppose that $\omega_1^X > \omega_1^{\mathrm{CK}}$. The odd column of V_*^X is a copy of $\mathcal{O}^X$, which computes both X and X'; thus $\mathrm{HJ}_{e_*}(X) \equiv_{\mathrm{T}} \mathcal{O}^X$. Spector's criterion gives $\mathcal{O} \leq_{\mathrm{HYP}} X$. Applying the implication $Y \leq_{\mathrm{HYP}} X \Longrightarrow \mathcal{O}^Y \leq_{\mathrm{T}} \mathcal{O}^X$ with $Y = \mathcal{O}$, we obtain

$$\mathcal{O}^{\mathcal{O}} \leq_{\mathrm{T}} \mathcal{O}^X \equiv_{\mathrm{T}} \mathrm{HJ}_{e_*}(X).$$

The two alternatives show that the stated degree interval is omitted. □

Corollary 2.4. *For the index e_*, if*

$$\mathcal{O} \leq_{\mathrm{T}} A <_{\mathrm{T}} \mathcal{O}^{\mathcal{O}},$$

then there is no real B such that

$$A \equiv_{\mathrm{T}} \mathrm{HJ}_{e_*}(B).$$

Consequently, Conjecture 4.5 of [1] is false. In fact, it remains false if its hypothesis $\mathcal{O} \leq_{\mathrm{T}} A$ is strengthened to $\mathcal{O} <_{\mathrm{T}} A$.

Proof. The first assertion follows immediately from Theorem 2.3. Moreover,

$$\mathcal{O} <_{\mathrm{T}} \mathcal{O}' <_{\mathrm{T}} \mathcal{O}^{\mathcal{O}}.$$

Here $\mathcal{O} <_{\mathrm{T}} \mathcal{O}'$ is the strictness of the Turing jump, while $\mathcal{O}' \leq_{\mathrm{T}} \mathcal{O}^{\mathcal{O}}$ follows from the uniform completeness of the hyperjump. If $\mathcal{O}^{\mathcal{O}} \leq_{\mathrm{T}} \mathcal{O}'$, then, since $\mathcal{O}' \leq_{\mathrm{HYP}} \mathcal{O}$, we would have $\mathcal{O}^{\mathcal{O}} \leq_{\mathrm{HYP}} \mathcal{O}$, contradicting $\mathcal{O} <_{\mathrm{HYP}} \mathcal{O}^{\mathcal{O}}$.

Thus, taking $A = \mathcal{O}'$ gives a counterexample with $\mathcal{O} <_{\mathrm{T}} A$, and in particular refutes Conjecture 4.5. □

Remark 2.5. The use of the even X'-column is not needed merely to refute Conjecture 4.5; it ensures the stronger conclusion $X <_{\mathrm{T}} \mathrm{HJ}_{e_*}(X)$ for every X. Thus the conjecture remains false even when restricted to pseudo-hyperjump operators that strictly increase the Turing degree of every input.

3. The class properties for pseudo-hyperjumps

We now turn to the consequences of the failure of Conjecture 4.5 for the associated class-characterisation questions. We first recall Properties 4.6–4.8 and then examine their logical relation to Conjecture 4.5.

Jananthan and Simpson ask for a characterisation of the Σ^1_1 classes $K \subseteq 2^{\mathbb{N}}$ satisfying the following three properties [1, Properties 4.6–4.8]. We state them explicitly, using $\emptyset$ in place of their 0. For columns we use their convention

$$(Z)_k = \{n : 2^k 3^n \in Z\}.$$

Property 4.6. Suppose $e \in \mathbb{N}$ and A is a real such that $\mathcal{O} \leq_{\mathrm{T}} A$. Then there exists $B \in K$ such that

$$A \equiv_{\mathrm{T}} \mathrm{HJ}_e(B) \equiv_{\mathrm{T}} B \oplus \mathcal{O}.$$

Property 4.7. Suppose $e \in \mathbb{N}$ and Z and A are reals such that

$$Z \oplus \mathcal{O} \leq_{\mathrm{T}} A \qquad \text{and} \qquad \emptyset <_{\mathrm{HYP}} (Z)_k \quad \text{for every } k \in \mathbb{N}.$$

Then there exists $B \in K$ such that

$$A \equiv_{\mathrm{T}} \mathrm{HJ}_e(B) \equiv_{\mathrm{T}} B \oplus \mathcal{O}$$

and

$$(Z)_k \not\leq_{\mathrm{HYP}} B \quad \text{for every } k \in \mathbb{N}.$$

Property 4.8. Suppose $e \in \mathbb{N}$ and Z and A are reals such that

$$Z \oplus \mathcal{O} \leq_{\mathrm{T}} A \qquad \text{and} \qquad \emptyset <_{\mathrm{HYP}} Z.$$

Then there exists $B \in K$ such that

$$A \equiv_{\mathrm{T}} \mathrm{HJ}_e(B) \equiv_{\mathrm{T}} B \oplus Z \equiv_{\mathrm{T}} B \oplus \mathcal{O}.$$

The logical relation between these properties and Conjecture 4.5 is worth making explicit. Each of Properties 4.6–4.8 implies Conjecture 4.5.

For Property 4.6 this is immediate by forgetting the requirement that the witness B belong to K. For Property 4.7, given $e \in \mathbb{N}$ and $A \geq_{\mathrm{T}} \mathcal{O}$, let

$$Z = \{2^k 3^n : k \in \mathbb{N},\ n \in \mathcal{O}\}.$$

Then $Z \equiv_{\mathrm{T}} \mathcal{O}$, every column $(Z)_k$ equals $\mathcal{O}$, and hence

$$Z \oplus \mathcal{O} \leq_{\mathrm{T}} A \qquad \text{and} \qquad \emptyset <_{\mathrm{HYP}} (Z)_k \quad \text{for every } k.$$

Thus Property 4.7 yields a $B \in K$ such that

$$A \equiv_{\mathrm{T}} \mathrm{HJ}_e(B) \equiv_{\mathrm{T}} B \oplus \mathcal{O}.$$

For Property 4.8, given $e \in \mathbb{N}$ and $A \geq_{\mathrm{T}} \mathcal{O}$, take $Z = \mathcal{O}$. Again its hypotheses are satisfied, and its conclusion contains

$$A \equiv_{\mathrm{T}} \mathrm{HJ}_e(B) \equiv_{\mathrm{T}} B \oplus \mathcal{O}.$$

Thus the observation of Jananthan and Simpson that, even if Conjecture 4.5 were true, the class-characterisation questions would remain open reflects an asymmetry between affirmative and negative answers: an affirmative answer to Conjecture 4.5 would not by itself characterise the Σ^1_1 classes satisfying Properties 4.6–4.8, whereas a negative answer forces each of the three collections of classes to be empty.

Corollary 3.1. *No class $K \subseteq 2^{\mathbb{N}}$, with or without a definability assumption, satisfies any one of Properties 4.6, 4.7, or 4.8 of* [1].

Proof. By the preceding observation, the existence of such a class for any one of Properties 4.6–4.8 would imply Conjecture 4.5, contradicting Corollary 2.4. □

4. Conclusion

We have shown that pseudo-hyperjump inversion fails at the level of Turing degrees. The failure is witnessed by a single pseudo-hyperjump HJ_{e_*} which strictly increases the Turing degree of every input, while its range omits every degree in the interval

$$\{\mathbf{d} : \deg_{\mathrm{T}}(\mathcal{O}) \leq \mathbf{d} < \deg_{\mathrm{T}}(\mathcal{O}^{\mathcal{O}})\}.$$

Thus the obstruction is stronger than the failure of inversion at the single target $\mathcal{O}$: since

$$\mathcal{O} <_{\mathrm{T}} \mathcal{O}' <_{\mathrm{T}} \mathcal{O}^{\mathcal{O}},$$

Conjecture 4.5 remains false even when its target is required to lie strictly above $\mathcal{O}$.

The construction is obtained by using the predicate $\omega_1^X = \omega_1^{\mathrm{CK}}$ as a uniform effective switch. On one side of the dichotomy the resulting pseudo-hyperjump has the Turing degree of X', while on the other it has the degree of $\mathcal{O}^X$. This produces the omitted interval while preserving strict Turing-degree increase. Finally, because each of Properties 4.6–4.8 implies Conjecture 4.5, the negative solution also settles the corresponding class-characterisation questions: no class of reals, with or without a definability assumption, satisfies any one of those properties.

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